

MediMind AI

An Agentic AI-Powered Multi-Agent Healthcare Companion

Concept Note | Lenovo LEAP: Generative AI & Agentic Systems Engineering Internship

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Department:	Mechanical Engineering	SDG 3 – Good Health & Well-being
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1. Introduction

Most consumer healthcare applications available today are limited to passive reminders or static data logging. They notify a patient at a fixed time, display a chart, or store a report as a file, but they do not reason about the patient's condition, plan a sequence of actions, or adapt when the patient deviates from a routine. This is a fundamental limitation for patients managing chronic conditions, elderly individuals living independently, or caregivers tracking multiple dependents, where healthcare management involves many interdependent tasks rather than isolated reminders.

Agentic AI addresses this gap by introducing systems composed of specialized agents that can reason about context, plan multi-step actions, retain memory across interactions, invoke external tools such as calendars and email, and execute tasks with minimal human input. MediMind AI applies this paradigm to healthcare management, coordinating multiple purpose-built agents through an orchestration layer rather than relying on a single monolithic chatbot. The objective is a system that behaves less like a reminder utility and more like an assistant capable of independent judgment within defined boundaries.

2. Problem Statement

- Patients, particularly the elderly and chronically ill, frequently miss medication doses due to lack of adaptive, persistent tracking.
- Appointments are forgotten or mismanaged because reminders are not linked to an agent capable of rescheduling or follow-up.
- Medical reports use clinical terminology that patients cannot interpret without assistance, delaying appropriate action.
- Existing apps monitor health data passively; they do not proactively flag risk patterns such as repeated missed doses.
- Emergency situations are detected late because no system continuously correlates medication, appointment, and vitals data to escalate risk.
- Applications such as Medisafe, Google Fit, and Apple Health are single-purpose: they track one dimension of health and do not coordinate decisions across medication, scheduling, reports, and lifestyle.

3. Objectives

- Design a multi-agent architecture in which each agent owns a distinct healthcare responsibility.
- Implement a Health Coordinator Agent that maintains patient context and delegates tasks to specialized agents.
- Build an n8n-based orchestration layer to execute agent workflows without a monolithic backend controller.
- Integrate OpenAI API for report interpretation, lifestyle recommendations, and agent-level reasoning.
- Automate medication tracking, appointment scheduling, and notification delivery through Google Calendar and Gmail APIs.
- Implement an emergency detection workflow that escalates to a caregiver when risk conditions are met.
- Deliver a deployed, responsive web application within a 6–7 day development window.

4. Proposed Solution

MediMind AI is structured as a Health Coordinator Agent supported by six specialized agents: Medication, Appointment, Report Analysis, Lifestyle, Emergency, and Notification. The Coordinator receives every user action or scheduled trigger, determines which specialized agent(s) should act, and passes the relevant context forward. Each specialized agent performs a narrow, well-defined task and reports its outcome back, which may trigger further agents — for example, a missed-medication event detected by the Medication Agent can trigger the Emergency Agent, which in turn triggers the Notification Agent to alert a caregiver.

This differs from a chatbot-based system in a specific way: a chatbot responds only when addressed and treats each query independently, with no persistent task ownership. MediMind AI's agents operate on schedules, webhook triggers, and inter-agent handoffs, executing actions autonomously in the background. The chatbot interface exists only as one entry point among several — most of the system's value is delivered without the user typing anything at all.

5. Why Agentic AI?

- **Reasoning:** OpenAI API is used by the Report Analysis and Lifestyle Agents to interpret unstructured medical text and produce patient-appropriate output.
- **Planning:** The Coordinator Agent sequences multi-step tasks, such as rescheduling an appointment and then notifying the patient and caregiver.

- **Memory:** Supabase stores persistent patient context — medication history, adherence patterns, and past reports — used by agents on every invocation.
- **Tool Calling:** Agents invoke Google Calendar, Gmail, and OpenAI as external tools through n8n nodes rather than hardcoded API calls.
- **Workflow Automation:** n8n executes each agent as a discrete workflow, triggered by webhooks, schedules, or outputs of other workflows.
- **Autonomous Execution:** Notifications, escalations, and rescheduling occur without a user actively initiating the action.
- **Multi-Agent Collaboration:** Agents hand off outcomes to one another, allowing one triggering event to produce a coordinated, multi-step response.

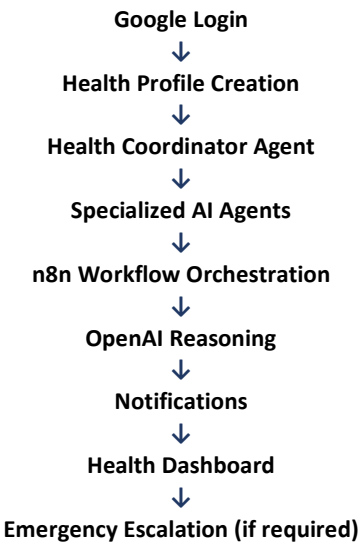
6. Key Features

- Medication reminders with adherence tracking and missed-dose detection
- Appointment scheduling with Google Calendar sync and automatic rescheduling
- Medical report upload and AI-based plain-language analysis
- Personalized diet, hydration, and activity suggestions
- Emergency detection with automatic caregiver escalation
- Email-based notifications for reminders, confirmations, and alerts
- Conversational AI assistant for direct health queries
- Health dashboard summarizing medication, appointments, and reports in one view

7. Technology Stack

Technology	Purpose
React.js	Frontend UI for patient dashboard, chat interface, and health profile
Tailwind CSS	Utility-first styling for responsive, mobile-friendly layout
Node.js	Backend runtime for API server and n8n webhook handlers
Express.js	REST API layer connecting frontend, Supabase, and n8n workflows
Supabase	Auth, PostgreSQL database, and storage for user, medication, and report data
OpenAI API	Reasoning engine for report analysis, lifestyle suggestions, and agent decisions
n8n	Workflow orchestration engine coordinating multi-agent execution
Google OAuth	Secure user authentication and calendar/gmail authorization
Google Calendar API	Appointment scheduling, rescheduling, and reminder sync
Gmail API	Automated email notifications and emergency alerts
GitHub	Version control and source code hosting
Vercel	Deployment and hosting of the frontend application

8. System Workflow



9. Multi-Agent Architecture

- **Health Coordinator Agent:** Master agent; maintains patient context, routes tasks to specialized agents, and makes delegation decisions.
- **Medication Agent:** Schedules doses, tracks adherence, triggers reminders, and flags repeated missed medication.

- **Appointment Agent:** Books and reschedules appointments via Google Calendar integration.
- **Report Analysis Agent:** Uses OpenAI to interpret uploaded reports, simplify terminology, and flag abnormal parameters.
- **Lifestyle Agent:** Generates diet, hydration, and exercise suggestions based on patient profile and report data.
- **Emergency Agent:** Monitors adherence and health signals for risk patterns and initiates caregiver escalation.
- **Notification Agent:** Dispatches email reminders, confirmations, and emergency alerts via Gmail API.

10. Innovation

MediMind AI is distinguished from existing consumer health platforms by coordinating decisions across multiple healthcare domains rather than tracking a single metric in isolation.

Platform	Core Limitation
Google Fit	Tracks fitness metrics passively; no medication management, no report analysis, no autonomous action
Samsung Health	Device-centric activity and vitals tracking; lacks reasoning, planning, or emergency escalation
Apple Health	Aggregates health data from apps and wearables; provides no active decision-making or agent coordination
Medisafe	Medication reminder app only; single-purpose, rule-based alerts with no reasoning or multi-domain coordination

MediMind AI differs by combining medication tracking, appointment management, report interpretation, and emergency escalation within one coordinated multi-agent system, where agents share context and trigger one another autonomously.

11. Expected Outcomes

- Improved medication adherence through persistent, context-aware reminders
- Reduced missed appointments via automated scheduling and rescheduling
- Faster patient comprehension of medical reports
- Earlier detection and escalation of emergency risk patterns
- Reduced manual caregiver monitoring effort
- A working demonstration of multi-agent orchestration using n8n and OpenAI

12. Future Scope

- Integration with wearable devices for continuous vitals monitoring
- Smartwatch support for real-time alerts
- Telemedicine integration for direct consultation scheduling
- Electronic Health Record (EHR) integration for longitudinal patient data
- IoT-based home health monitoring
- Predictive healthcare analytics using historical adherence and vitals data
- Voice assistant interface for hands-free interaction

13. SDG Alignment

MediMind AI aligns with SDG 3 (Good Health and Well-being) by improving medication adherence, enabling timely medical report comprehension, and reducing emergency response delays. By automating routine healthcare coordination, the system reduces dependence on constant caregiver supervision and improves healthcare accessibility for elderly and chronically ill patients managing care independently.

14. Conclusion

MediMind AI demonstrates Agentic AI principles applied to a practical healthcare use case: multiple specialized agents, coordinated through n8n, reason over patient context, plan multi-step actions, and execute them with minimal manual intervention. The architecture is deliberately scoped for a 6–7 day build using tools that integrate directly — Supabase, OpenAI, n8n, and Google APIs — while remaining extensible toward wearables, EHR systems, and predictive analytics. The result is a system that goes beyond reminder-based healthcare apps by demonstrating genuine autonomous, multi-agent decision-making.

15. GitHub

[GitHub Repository: \(To be added after development\)](#)

16. Live Demo

Live Demo: (To be updated after deployment)